



Barcodes for product information

reduces fraud risk and operational disruption, fostering trust and safety.

User experience, sustainability, and long-term value

The integration of EAS and RFID into T-shirts also enhances the user experience. RFID's contactless scanning enables frictionless self-checkout in retail environments, reducing wait times and improving customer satisfaction. In events or workplaces, RFID tags allow seamless access to rooms or restricted zones without physical keycards or paper tickets.

RFID technology further enables interactive and personalised experiences. Conference attendees may access personalised schedules, networking opportunities, and exclusive content via linked applications activated by RFID-enabled apparel. Concertgoers may receive tailored updates or VIP access, while educational institutions can connect students to campus services or learning resources via wearable IDs.

Many organisations across retail and institutional sectors have adopted printed RFID and EAS technologies with measurable outcomes. Retailers report theft reductions of up to 70% while achieving

inventory accuracy levels approaching 99%. These gains translate into cost savings, improved product availability, and stronger customer loyalty. Educational institutions using RFID-enabled garments have streamlined attendance management, reduced administrative costs, and enhanced data reliability, enabling earlier academic interventions and more targeted student support. Events deploying RFID for access and crowd management report smoother entry processes, enhanced safety, and an improved attendee experience.

Beyond immediate operational benefits, integrating EAS and printed RFID into T-shirts supports sustainability and regulatory compliance. The emergence of digital product passports, which provide verifiable product data linked to RFID tags, is expected to become mandatory under regulations such as the EU's Ecodesign for Sustainable Products Regulation (ESPR). These digital records support circular economy initiatives by enabling traceability throughout the garment lifecycle, from manufacturing and retail to resale, repair, and recycling. Durable RFID tags allow garments to remain traceable for many years, facilitating

responsible end-of-life processing and reducing textile waste. This technological convergence helps brands meet growing consumer demand for transparency and sustainability while complying with evolving legislation.

Comfort and durability remain key considerations in wearable technology integration. Printed RFID tags have evolved to be soft, flexible, and resistant to repeated wash cycles, with some capable of withstanding over 100 wash cycles without performance degradation. Innovative approaches include thread-based RFID solutions, such as E-Thread, embedded directly into the textile weave, and premium printable inlays that preserve garment comfort and appearance. The tags are discreet and often invisible, ensuring they do not affect fit, feel, or style.

Some RFID tags are also designed to coexist with barcodes and QR codes, enabling multi-layered product information and branding opportunities. These advances make RFID and EAS integration a practical and scalable solution for fashion manufacturers and other sectors seeking to unlock the full potential of wearable technology.

Final thoughts

The integration of electronic article surveillance and printed RFID technology in T-shirts represents a significant advancement across security, identity management, inventory efficiency, and sustainability. Whether deployed in retail spaces, corporate campuses, concerts, or educational institutions, RFID-embedded apparel bridges physical and digital environments, streamlining operations and enhancing user experience. As technology costs decline and system interoperability improves, this convergence of EAS and RFID is positioned to become a foundational element for future-ready brands, event organisers, and institutions seeking to strengthen security, authenticity, and operational intelligence while supporting a more sustainable and connected world. **EFY**